

# Customer Churn Prediction – Project Documentation

## 1. Project Overview

This project focuses on analyzing customer churn and building predictive models to identify customers likely to discontinue services. The objective is to uncover key churn drivers and provide data-driven business recommendations to improve customer retention.

## 2. Methodology & Steps Performed

- Data loading and initial inspection
- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA) with visualizations
- Feature engineering and encoding
- Train-test split with stratification
- Model training using multiple classifiers
- Model evaluation using Accuracy, Precision, Recall, F1-score, and ROC-AUC
- Confusion matrix visualization
- Feature importance analysis (where applicable)

## 3. Key Findings & Conclusions

The analysis shows that churn is significantly higher among new customers and those on month-to-month contracts. Limited service adoption and manual payment methods further increase churn risk. Customers with long-term contracts and bundled services demonstrate stronger retention.

## 4. Business Impact

The predictive model helps businesses proactively identify high-risk customers, enabling targeted retention strategies such as personalized onboarding, service bundling, and contract upgrades.

## 5. Future Scope & Improvements

- Hyperparameter tuning for improved model performance
- Handling class imbalance using advanced techniques
- Deployment of the model as a web application or API
- Real-time churn prediction using streaming data
- Integration of customer feedback and behavioral data